



Hoppers Challenge: Coding for Sustainability

A sustainable world is one in which our environmental systems and societies flourish, natural resources are able to replenish themselves, and people are able to live healthy and secure lives. The growing complexity and scale of scientific computing workflows and AI training pipelines place huge demands on energy and other natural resources. Sustainability can be overlooked in favour of technological advancements, therefore it is of utmost importance for us to make the effort to consider how we can fit sustainability into our own projects.

The United Nation's Sustainable Development Goals:

<https://sustainability.ed.ac.uk/governance-publications-reports/reports/2021-22/sustainable-development-goals>

Report archives: <https://uoefinance.ed.ac.uk/annual-report-and-accounts-2024-25/archive>

The Challenge

Take one of the University's initiatives or actions from their report archives or their sustainability website, and design an interactive educational tool to educate students, staff and others on it. Think of how you can make it more engaging, accessible and meaningful.

Sustainable Development Goals

(You need to check the website for more information)

1. No poverty
2. Zero hunger
3. Good health and wellbeing
4. Quality education
5. Gender equality
6. Clean water and sanitation
7. Affordable and clean energy
8. Decent work and economic growth
9. Industry, innovation and infrastructure
10. Reduced inequalities
11. Sustainable cities and communities
12. Responsible consumption and production
13. Climate action
14. Life below water



- 15. Life on land
- 16. Peace, justice and strong institutions
- 17. Partnerships for the goals

Beginner Section

Stuck for ideas or not sure what to do? Have a look at the following websites to get started.

HTML Tutorial (structure of a webpage):

<https://www.w3schools.com/html/>

CSS Tutorial (styling of a webpage):

<https://developer.mozilla.org/en-US/docs/Web/CSS/Tutorials>

How do games work in Python? <https://www.pythoncentral.io/how-to-develop-a-game-with-python/>

Pong in Python:

<https://www.freecodecamp.org/news/how-to-code-pong-in-python/>

Flappy Bird in Python:

<https://www.geeksforgeeks.org/python/how-to-make-flappy-bird-game-in-pygame/>

Snake in Java:

<https://www.educative.io/blog/javascript-snake-game-tutorial>

Simple Quiz in Java:

<https://www.geeksforgeeks.org/java/online-quiz-system-in-java/>



How will we judge?

Marks will be given for the following:

- Teamwork (how much did everyone contribute)
- Originality
- Theme (How well does your project fit the theme of Sustainability?)
- Presentation Skills
- Project Quality

Good Luck!